Mathematical Statistics

Lecture 10 Expectation, Variance, and Moments

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Contents

1 Lecture Overview

This lecture covers three fundamental concepts in the study of random variables:

1. **Mathematical Expectation:** The formal definition of the average or mean of a random variable.
2. **Variance and Standard Deviation:** Measures of how spread out or dispersed a distribution is from its mean.
3. **Moments:** A generalized concept that describes the shape of a distribution, with expectation and variance being special cases.

We will explore these concepts for both **discrete** and **continuous** random variables and solve a variety of problems to solidify understanding. These tools are essential for analyzing all major probability distributions (Binomial, Poisson, Normal, etc.).

2 Part 1: Mathematical Expectation (The Mean)

2.1 Introduction to Expectation

Mathematical Expectation, denoted as $\boldsymbol{E[X]}$, represents the long-run average value of a random variable X. If we were to repeat a random experiment an infinite number of times, the average of the outcomes would converge to the expected value.

Why This Matters: Center of Mass

Expectation is synonymous with the **mean** (μ or $\bar{X}$). It tells us the "center of mass" of a probability distribution, providing a single value that summarizes the central tendency of the data.

2.2 Expectation for a Discrete Random Variable

For a discrete random variable X that can take values $x_1, x_2, \ldots, x_n$ with corresponding probabilities $p_1, p_2, \ldots, p_n$, the expected value is a weighted average.

Formula for Discrete Expectation

$$E[X] = \sum_{i=1}^{n} (x_i \cdot p_i)$$

2.2.1 Example 1: Tossing a Coin Twice

Problem: Let X be the number of heads in two fair coin tosses. Find $E[X]$.
Solution:

1. **Sample Space:** {HH, HT, TH, TT}.
2. **Random Variable & Probabilities:**
   - $P(X = 0) = P(\text{TT}) = 1/4$
   - $P(X = 1) = P(\text{HT or TH}) = 2/4$

- $P(X = 2) = P(\text{HH}) = 1/4$

3. **Calculate E[X]:**

$$E[X] = \left(0 \cdot \frac{1}{4}\right) + \left(1 \cdot \frac{2}{4}\right) + \left(2 \cdot \frac{1}{4}\right) = 0 + \frac{2}{4} + \frac{2}{4} = \frac{4}{4} = 1$$

Conclusion: On average, we expect to get **1 head**.

2.3 Expectation for a Continuous Random Variable

For a continuous random variable with a PDF $f(x)$, the expectation is found by integrating the product of x and $f(x)$ over its entire range.

Formula for Continuous Expectation

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

2.4 Properties of Expectation

- **Linearity:** $E[aX + b] = aE[X] + b$, where a and b are constants.
- **Additivity:** $E[X + Y] = E[X] + E[Y]$.
- **Multiplication (Independent Variables):** $E[XY] = E[X] \cdot E[Y]$.

3 Part 2: Variance and Standard Deviation

3.1 Introduction to Variance

While expectation gives the center, **variance** measures the **spread** or **dispersion** of a distribution. A small variance means data points are clustered tightly around the mean; a large variance means they are spread out.

Core Concept: Variance

Variance is the **expected squared deviation from the mean**.

- **Definitional Formula:** $\text{Var}(X) = \sigma^2 = E[(X-\mu)^2]$, where $\mu = E[X]$.
- **Computational Formula:** $\text{Var}(\boldsymbol{X}) = \boldsymbol{E[X^2] - (E[X])^2}$.

To use the computational formula, we first need to calculate $E[X^2]$.

- **Discrete:** $E[X^2] = \sum(x_i^2 \cdot p_i)$
- **Continuous:** $E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot f(x)\,dx$

3.1.1 Example 2: Variance of a Die Roll

Problem: Calculate the variance for a single roll of a fair die.
Solution:

1. **Find** $E[X]$**:** From earlier examples, $E[X] = 3.5$.
2. **Find** $E[X^2]$**:**

$$\begin{aligned} E[X^2] &= \left(1^2 \cdot \frac{1}{6}\right) + \left(2^2 \cdot \frac{1}{6}\right) + \cdots + \left(6^2 \cdot \frac{1}{6}\right) \\ &= \frac{1}{6}(1+4+9+16+25+36) = \frac{91}{6} \end{aligned}$$

3. **Calculate Variance:**

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12} \approx 2.917$$

3.2 Standard Deviation

The **Standard Deviation** (σ) is the square root of the variance. It is preferred for interpretation as it is in the same units as the mean.

$$\sigma = \sqrt{\text{Var}(X)}$$

3.3 Properties of Variance

- $\text{Var}(X) \geq 0$.
- $\text{Var}(b) = 0$ for a constant b.
- $\text{Var}(aX+b) = a^2\text{Var}(X)$. (The constant b is dropped, and a is squared).
- $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$ for independent variables.

4 Part 3: Moments

4.1 Introduction to Moments

Moments are statistical parameters that describe the shape of a distribution. Expectation and variance are the two most important moments.

Types of Moments

1. **Moments about the Origin (Raw Moments):** Describe the distribution relative to zero. Denoted by μ'_r.
$$\mu'_r = E[X^r]$$
   - $\mu'_1 = E[X]$ is the **mean**.
   - $\mu'_2 = E[X^2]$.
2. **Moments about the Mean (Central Moments):** Describe the shape relative to the mean μ. Denoted by μ_r.
$$\mu_r = E[(X - \mu)^r]$$
   - $\mu_1 = E[X - \mu] = 0$ (always).
   - $\mu_2 = E[(X - \mu)^2]$ is the **variance**.
   - μ_3 measures **skewness** (asymmetry).
   - μ_4 measures **kurtosis** (tailedness).
3. **Moments about an Arbitrary Point 'a':** The most general form.
$$E[(X - a)^r]$$

4.2 Relationships Between Moments

Central moments can be expressed in terms of raw moments.

- $\mu_2 = \mu'_2 - (\mu'_1)^2$
- $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$
- $\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$

4.2.1 Example 3: Finding Moments

Problem: The first four moments of a distribution about the point $a = 4$ are $-1.5, 17, -30,$ and 108. Find the first three raw moments (moments about the origin).

Solution: We are given:

- $E[X - 4] = -1.5$
- $E[(X - 4)^2] = 17$
- $E[(X - 4)^3] = -30$

1. **Find $\mu'_1 = E[X]$:**
$$E[X - 4] = E[X] - 4 = -1.5$$
$$E[X] = 4 - 1.5 = 2.5 \implies \boldsymbol{\mu'_1 = 2.5}$$

2. **Find $\mu_2' = E[X^2]$:**

$$\begin{aligned} E[(X-4)^2] &= E[X^2 - 8X + 16] = 17 \\ E[X^2] - 8E[X] + 16 &= 17 \\ E[X^2] - 8(2.5) + 16 &= 17 \\ E[X^2] - 20 + 16 = 17 &\implies E[X^2] = 21 \implies \boldsymbol{\mu_2' = 21} \end{aligned}$$

3. **Find $\mu_3' = E[X^3]$:**

$$\begin{aligned} E[(X-4)^3] &= E[X^3 - 12X^2 + 48X - 64] = -30 \\ E[X^3] - 12E[X^2] + 48E[X] - 64 &= -30 \\ E[X^3] - 12(21) + 48(2.5) - 64 &= -30 \\ E[X^3] - 252 + 120 - 64 &= -30 \\ E[X^3] - 196 = -30 &\implies E[X^3] = 166 \implies \boldsymbol{\mu_3' = 166} \end{aligned}$$

5 Lecture Summary

Key Takeaways

- **Expectation ($E[X]$)** is the mean or average of a random variable.
- **Variance (Var(X))** measures the spread of the distribution from its mean. The computational formula $\boldsymbol{E[X^2] - (E[X])^2}$ is key.
- **Moments** provide a systematic way to describe the shape of a distribution.
  - **Raw Moments** ($\mu_r' = E[X^r]$) are about the origin. The 1st raw moment is the mean.
  - **Central Moments** ($\mu_r = E[(X-\mu)^r]$) are about the mean. The 2nd central moment is the variance.

6 Practice Problems

1. **Discrete Variable:** The number of calls received by a call center in an hour follows the distribution below. Find the expected number of calls and the variance.

x **(Calls)**	0	1	2	3
$p(x)$	0.1	0.3	0.4	0.2

2. **Continuous Variable:** A random variable X has the PDF $f(x) = \frac{1}{8}x$ for $0 \le x \le 4$.
   (a) Find the expected value, $E[X]$.
   (b) Find the variance, Var(X).

3. **Moments:** The first three raw moments (μ_1', μ_2', μ_3') of a distribution are $2, 8$, and 30, respectively.
   (a) Find the mean and variance of the distribution.
   (b) Find the third central moment, μ_3, which describes the skewness.